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Theoretical Adsorption Models for the Methane–Shale System

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Abstract

Gas sorption behavior needs to be understood very well to quantify the reservoir potential of unconventional reservoirs. To quantify adsorption capacity of rock samples, experiments are performed for a wide range of pressure and temperature with relevant fluid. By utilizing experimental adsorption data, adsorption isotherms are deployed to quantify the amount of gas adsorbed into a rock and its equilibrium state. Currently, this study is considering the adsorption data of methane (CH₄)-shale system from literature databases under various pressures and specific temperatures, including moisture, compressional and non-compression conditions. This study aims to assess the maximum CH₄ adsorption capacity and the effect of deformation on the selectivity of isotherms equilibrium constants. More specifically, a modeling approach was used to evaluate the suitability of Langmuir, Toth, and Henry isotherms for a methane/shale system using experimental adsorption data. The key conceptual novelty of this research lies in evaluating the predictive power of traditionally accepted gas adsorption isotherms on or against methane/shale adsorption data, specifically up to 15 MPa but at 30°C and moisture conditions. Results obtained by ranking of isotherms produced as Langmuir = Toth > Henry based on RMSE and R² for both samples, including no effect of rock deformation on selectivity of isotherms.

Introduction

Methane has been regarded as a very crucial element of the world's energy scenario (Wuebbles and Hayhoe, 2002), and it is also the second most important candidate for potential greenhouse gas (GHG) (Deng et al., 2024, Sawyer et al., 2022) after carbon dioxide. Unconventional gas reservoirs, particularly shale, are categorized based on accumulation mechanisms (Jin et al., 2022). In particular, stimulation is required in shale gas reservoirs to economically produce these reservoirs (Sunjay and Kothari, 2011). Conceptually, rock's matrix is comprised of inorganic detritus and kerogen (Glorioso and Rattia, 2012). The low permeability and high porosity shale rock comprises a large proportion in organic matter, clay, silt, which mainly develop the anisotropic and heterogeneous conditions of the shale (Raza et al., 2023). Gas transport in shale formations occurs through pores, along rock surfaces, and within fluids. For example, gas molecules reside in the porous space of rocks, attach to the walls of those pores or dissolve in both kerogen and clay present in the rocks (Sunjay and Kothari, 2011, Alafnan et al., 2021). Shale is characterized

by extremely small pores that are densely distributed, along with secondary heterogeneities introduced by fissures and microfractures (Figure 1) (Wang and Fidelibus, 2021).

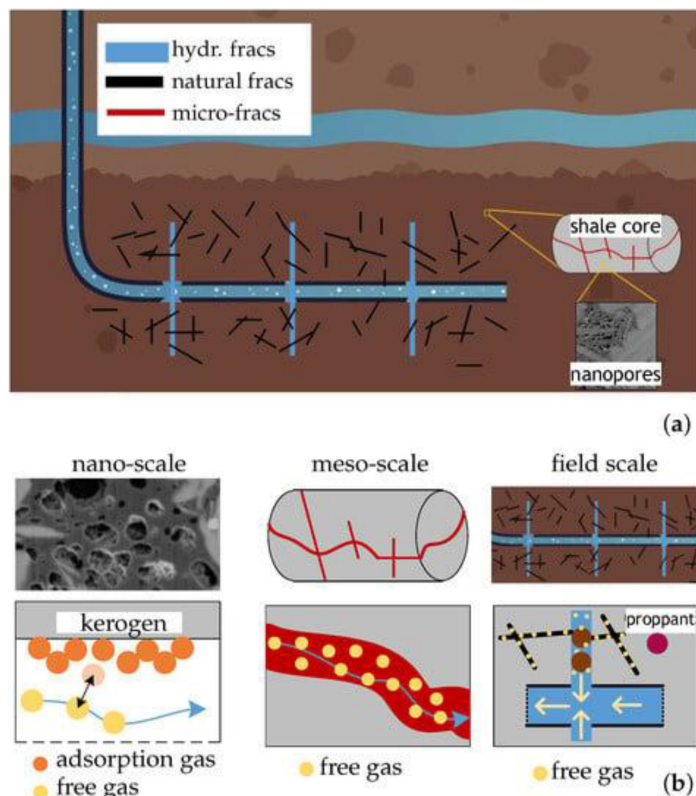


Figure 1—Shale gas transport mechanisms as free, dissolved, adsorbed gas from pores, rock surface and organic regions (Wang and Fidelibus, 2021).

Shale contains gas in free, adsorbed and dissolved phases (Alafnan et al., 2021), thus the recovered gas is comprised of not only free gas but also desorbed gas from the surface of the rock. A significant portion of the total gas in place is typically present in the form of adsorbed gas (Tian et al., 2019). Various factors such as pressure, temperature, salinity, TOC, kerogen maturity, surface area, etc affect the adsorption of a rock (Klewiah et al., 2020). Gas adsorption can be quantified on an absolute basis using molecular simulation (Glatz et al., 2022b, Glatz et al., 2022a) or nuclear magnetic resonance (Banas et al., 2005), while excess adsorption is typically estimated using volumetric, chromatographic, and gravimetric methods (Raza, 2025). Under specific temperature conditions, isotherms are used to describe the amount of maximum gas adsorbed at infinite equilibrium pressure (Vardcharragosad and Ayala H, 2015). The applicability of isotherms depends on the rock-fluid system as unconventional reservoirs typically accept Type I Langmuir isotherms (Alafnan et al., 2021). Understanding gas adsorption requires knowledge of both the amount of gas adsorbed onto the rock surface and the selectivity of the adsorption model (Alafnan et al., 2021).

Fitting experimental data to adsorption isotherm models enables the determination of key parameters required for modeling and predicting gas behavior in shale reservoirs. This study aims to theoretically assess CH_4 adsorption on shale using representative models (Langmuir, Toth, and Henry isotherms) to evaluate their suitability based on experimental adsorption data.

Methodology

From the published methane adsorption data by Liang et al. (2021), a dataset comprising two shale samples was extracted for analysis under both original (O) and compressively deformed (D) conditions,

within a pressure range of 0–15 MPa at 30°C. Methane adsorption was measured using high-pressure sorption experiments conducted under moisture equilibrium conditions. The relative humidity during these measurements was controlled using saturated K_2SO_4 solutions in an evacuated dryer. Continuous monitoring indicated a moisture content ranging from 1.26% to 1.68% in the samples.

Figure 2 presents adsorption (mmol/g) vs. pressure (MPa) for the two shale samples, S3 and S4. The labels ‘O’ and ‘D’ indicate the original (undeformed) and deformed conditions of each sample, respectively. Adsorption increases with pressure in all cases but begins to plateau beyond ~8 MPa. The S4 samples (both original and deformed) exhibit higher adsorption than the S3 samples, indicating a greater adsorption capacity. Deformation reduces adsorption in both shale types, implying that structural changes obstruct gas uptake.

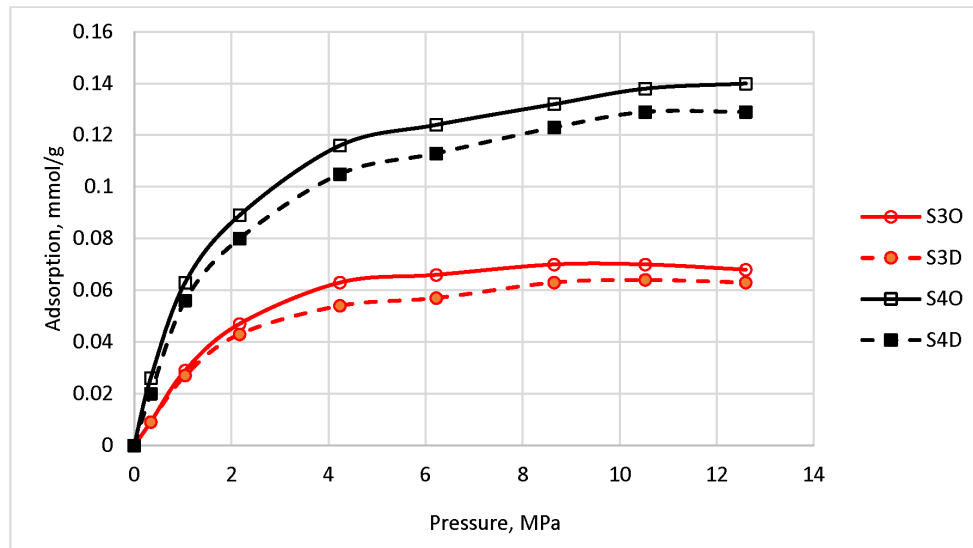


Figure 2—Adsorption data of two shale samples (S3 and S4) under original (non-deformation, O) and compressive (deformation, D), versus pressure.

This section briefly describes the adsorption models that have been used for methane/shale adsorption predictions. The key aspect in assessing these models is determining their ability to accurately predict the adsorption behavior of individual components. The coefficient of determination (R^2) is generally used to evaluate the accuracy of adsorption models and has been widely emphasized in previous studies (Wang et al., 2016, Ho, 2006, Wang et al., 2020). R^2 indicates the proportion of variance in the dependent variable that is explained by the regression model. Its value ranges from 0 to 1, where 1 signifies a perfect fit and 0 indicates that the model explains none of the variance. The adjusted R^2 describes the significance of variance, accounting for the number of variables, and it can never hold a greater value than R^2 . Mean Square Error (MSE) is the variance of the residuals, whereas Root Mean Square Error (RMSE), or the square root of MSE, is the standard deviation of the residuals. The RMSE is often preferred over MSE because it retains the same units as the dependent variable, allowing for more intuitive interpretation and direct comparison. These metrics will be used to rigorously evaluate the performance of the adsorption models in representing methane adsorption behavior in shale.

$$R^2 = 1 - \frac{\sum (Truevalues - Predictedvalues)^2}{\sum (Truevalues - Averagevalues)^2} \quad (1)$$

$$R^2_{adj} = 1 - \left[\frac{(1 - R^2)(no.of\ observations - 1)}{(no.of\ observations - no.of\ independentvariables - 1)} \right] \quad (2)$$

$$MSE = \frac{1}{N} \sum_{i=1}^N (\text{truevalues} - \text{predictedvalues})^2 \quad (3)$$

$$RMSE = \sqrt{MSE} = \sqrt{\frac{1}{N} \sum_{i=1}^N (\text{truevalues} - \text{predictedvalues})^2} \quad (4)$$

Langmuir

The work of Langmuir led to a model that provides an effective explanation for adsorption ([Langmuir, 1916](#), [Langmuir, 1917](#), [Langmuir, 1918](#)). The Langmuir isotherm is a classical model that often provides the best fit to experimental adsorption data. It states that the first layer of adsorbate is held at the surface during adsorption ([Raza et al., 2023](#)).

$$q = \frac{q_L bP}{bP + 1} \quad (5)$$

In the Langmuir model, the quantity of adsorbed molecules onto the rock, denoted as q (in mmol/g), is related to the equilibrium pressure P (in MPa). It should be noted that q_L is the Langmuir volume, anticipated quantity of adsorbed molecules as the equilibrium pressure approaches infinity ($P \rightarrow \infty$), [Eq. 5](#) yields q_L . The determination of q_L and b (inversely proportional to Langmuir pressure) typically involves non-linear regression, where optimization is performed based on RMSE.

Tóth

Toth model is a modified form of the Langmuir model, with improved accuracy of the predicted values to the laboratory adsorption results and covering the sub-monolayer adsorption phenomenon. It applies to low and high adsorbate concentrations as well as heterogeneous surfaces ([Serafin and Dziejarski, 2023](#)). The mathematical form of this model is given in [Eq. 6](#). ([Glatz et al., 2022b](#))

$$q = q_\infty \times \left[\frac{(bP)^{\frac{1}{n}}}{(bP)^{\frac{1}{n}} + 1} \right]^{m \times n} \quad (6)$$

where q_∞ is adsorbed gas molecule at infinite pressure. [Eq. 6](#) can be written in another version ([Kumar et al., 2011](#), [Tóth, 1995](#)):

$$q = \frac{q_T P}{(b + P^t)^{\frac{1}{t}}} \quad (7)$$

where b , t , and q_T Are determined on an individual basis using [Eq.\(7\)](#).

Henry

Henry's isotherm ([Kalam et al., 2021](#)) is a simple, one-parameter adsorption model that describes a linear relationship between the amount of adsorbate adsorbed and its equilibrium concentration in solution. The relationship is given by:

$$q_e = K_{HE} C_e \quad (8)$$

where, q_e is the amount of adsorbate adsorbed at equilibrium (mmol/g), K_{HE} is Henry's adsorption constant (L/g), C_e is the equilibrium concentration of the adsorbate in solution (mg/L). For gases, C_e is replaced by equilibrium pressure (usually in MPa or bar). Plotting q_e against C_e yields a straight line, with the slope equal to K_{HE} . Henry's model is most applicable when the adsorbent surface coverage is very low, meaning very few adsorption sites are occupied. It accurately reflects the adsorption trend at low concentrations of solute, demonstrating monolayer adsorption behavior in dilute systems.

Results and Discussions

Langmuir Model

The purpose was to study how changes in pressure and temperature influence the depiction of gas-solid systems based on the Langmuir parameters (as shown in Figure 3). According to the results, the Langmuir model provided a good fit for predicting the amount of methane absorbed on the core sample. Since the RMSE is zero, we can say that the model's predictions are reliable. However, a slight discrepancy remained between the model trend and the experimental data, suggesting the influence of additional factors. As the temperature increased at low pressure, the amount of gas adsorbed decreased. This observation in the Langmuir model indicates S3O has a capacity of 0.1 mmol/g, whereas S3D, S4O, and S4D have capacities of the same level (0.1 mmol/g, 0.2 mmol/g, and 0.1 mmol/g, respectively). The lessening of S4 is a result of deformation behavior, as seen in experiments. Both samples led to an RMSE of 0.00, showing that they matched the observed data well. This means the adsorption capacity estimates for both samples are highly precise. These findings align with the established understanding that unconventional reservoirs typically follow Type I Langmuir isotherms (Alafnan et al., 2021). With these discoveries, scientists can now design effective models for gas-solid adsorption by taking factors like pressure, temperature, and Langmuir parameters into account. Results are presented in Table 1.

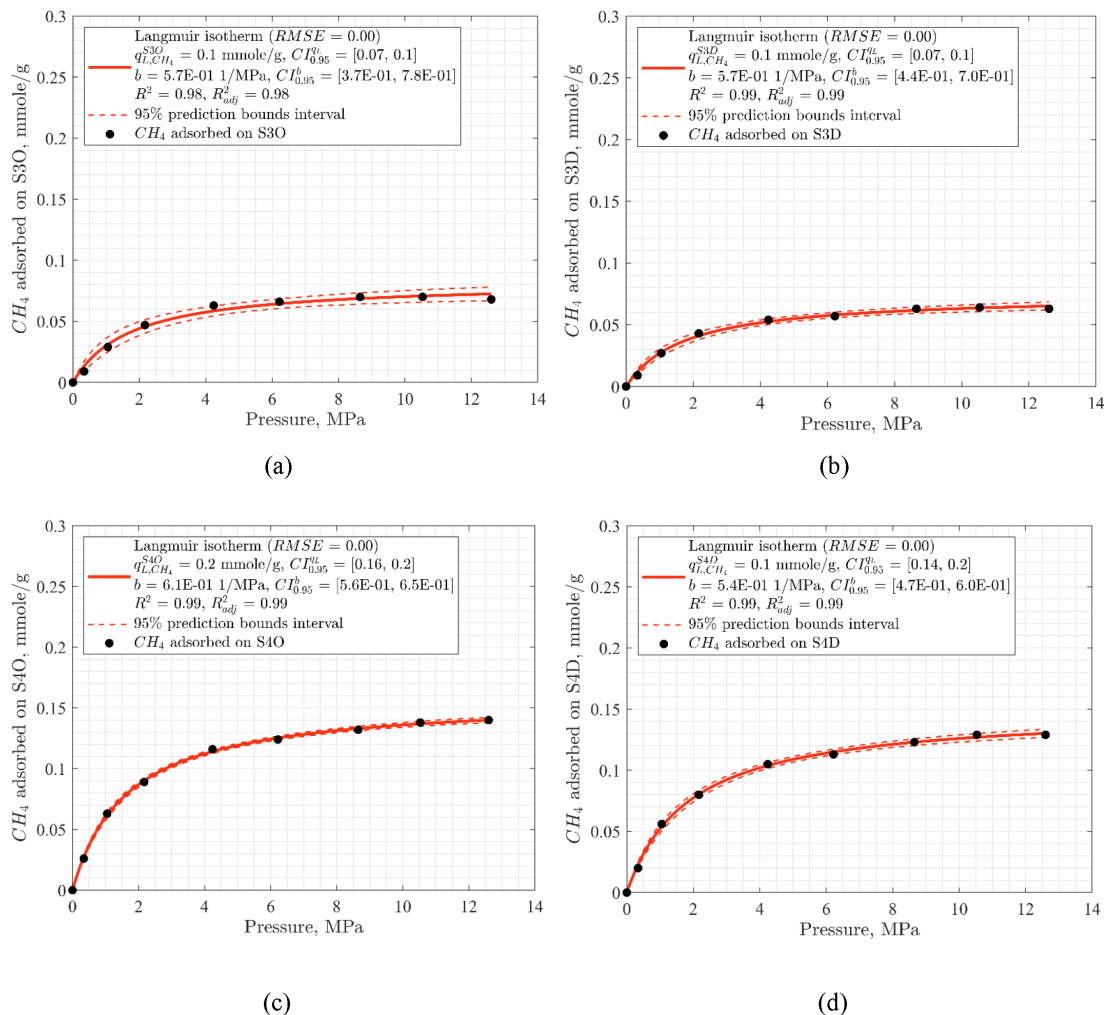


Figure 3—Langmuir adsorption isotherm trends of methane/shale system under original (non-deformation, O) and compressive (deformation, D) versus pressure for samples: (a) S3O; (b) S3D; (c) S4O; (d) S4D

Table 1—Langmuir isotherms parameters after fitting with experimental data for the shale system

Parameter	S3O	S3D	S4O	S4D
$q_{L,mmole/g}$	0.1	0.1	0.2	0.1
Reduced volume (mmol/g rock)	0		0.1	
Reduced rate, %	0		50	
$b, l/MPa$	5.7E-01	5.7E-01	6.1E-01	5.4E-01
RMSE	0	0	0	0
R^2	0.98	0.99	0.99	0.99
R_{adj}^2	0.98	0.99	0.99	0.99

Figure 4 illustrates the Toth isotherm for methane (CH_4) adsorption on the S3O sample. The experimental results (black dots) show a strong correlation with the Toth equation (red curve), as seen from the R^2 value of 0.99. The red dashed lines indicate the 95% prediction range. The adsorption capacity reaches 0.1 mmol/g, with parameters $b = 1.7$ and $t = 1.00$, indicating favorable adsorption behavior. The model is an excellent match for the data, as its RMSE is 0.00. Since the Toth isotherm uses three parameters (q_T , b and t) instead of two (q_T , b) like the Langmuir model. Toth volume remains the same in S3 sample, while decreases in S4D. RMSE is found zero and R^2 up to 0.99 in all cases except S3D. Importantly, b increased in the deformation case in both samples.

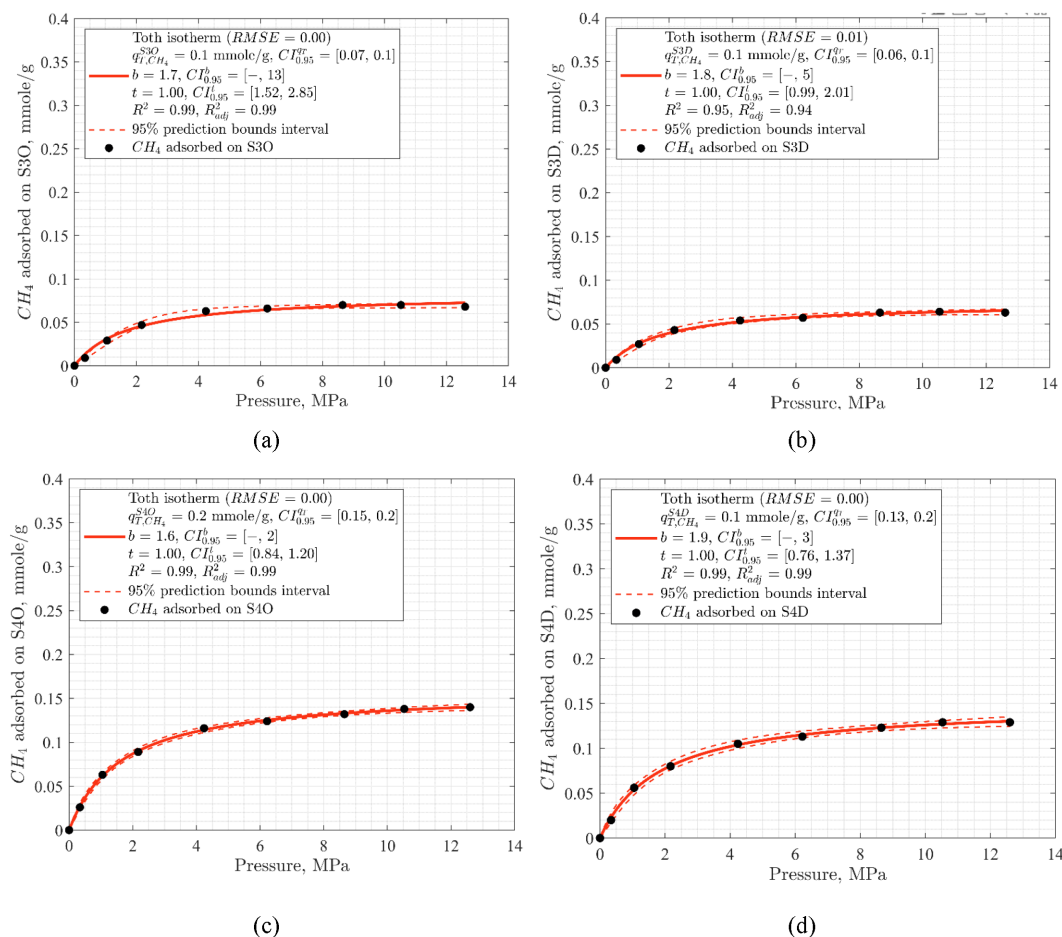


Figure 4—Toth isotherm trends of methane/shale system under original (non-deformation, O) and compressive (deformation, D), versus pressure for samples: (a) S3O; (b) S3D; (c) S4O; (d) S4D

Table 2—Toth isotherms parameters after fitting with experimental data for the shale system

Parameter	S3O	S3D	S4O	S4D
$q_{T,mmole/g}^{S3O}$	0.1	0.1	0.2	0.1
Reduced volume (mmol/g rock)	0		0.1	
Reduced rate, %	0		50	
$b, 1/MPa$	1.7	1.8	1.6	1.9
RMSE	0	0.01	0	0
R_2	0.99	0.95	0.99	0.99
R_{adj}^2	0.99	0.94	0.99	0.99

Henry Model

Figure 5 shows the Henry adsorption isotherms for the shale samples in their original and deformed states. Although deformation generally does not favor methane adsorption, the S3D and S4D samples adsorb more methane than their original counterparts. This suggests that deformation may alter adsorption mechanisms in shale, possibly through the formation of microfractures and enhanced fluid flow. Adsorption increases sharply, reaching a peak and then stabilizing at pressures below 2 MPa. The R^2 values ranging from 0.41 to 0.49 and RMSE from 0.02 to 0.04, indicate a poor fit between the model and the experimental data. Slight increases in the Henry coefficient for both samples were observed, probably signifying their ability to adsorb more methane. The paper claims that geomechanical deformation is an important factor in the storage of gases in shale.

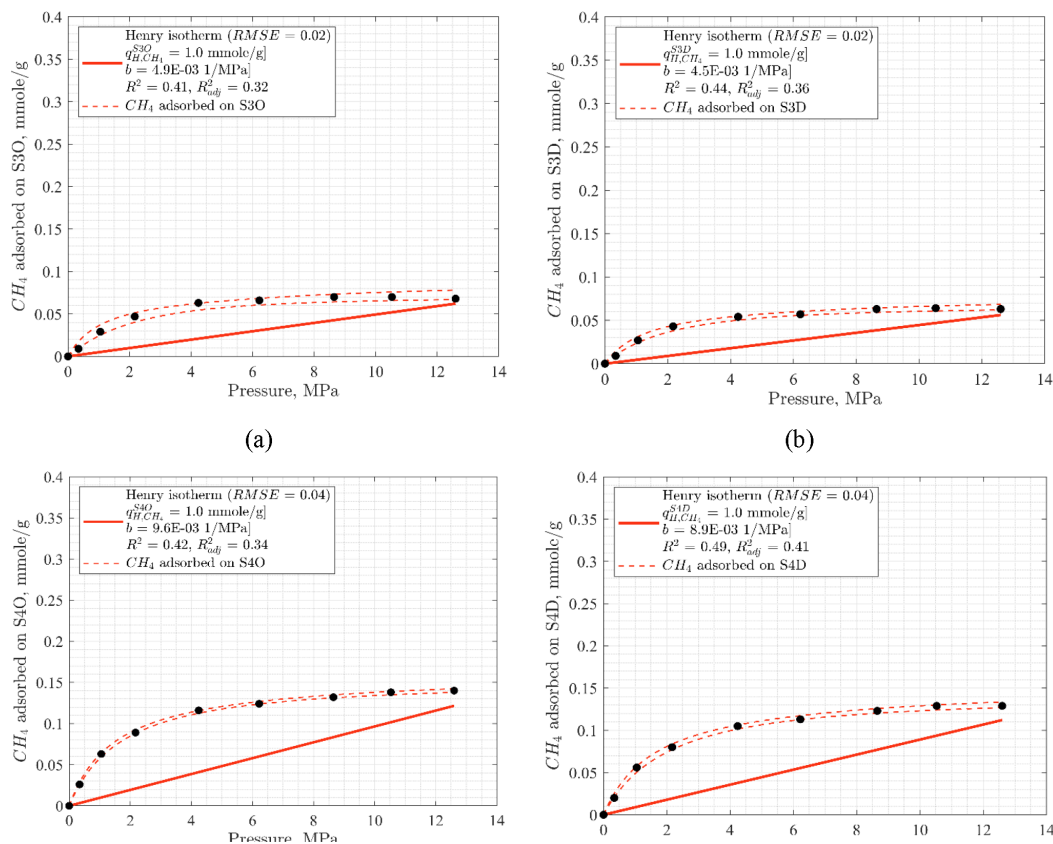


Figure 5—Henry isotherm trends of methane/shale system under original (non-deformation, O) and compressive (deformation, D), versus pressure for samples: (a) S3O; (b) S3D; (c) S4O; (d) S4D

Table 3—Henry isotherm parameters after fitting with experimental data for the shale system

Parameter	S3O	S3D	S4O	S4D
$q_{H,mmole/g}$	1	1	1	1
Reduced volume (mmol/g rock)	0		0	
Reduced rate, %	0		0	
$b, l/MPa$	4.9E-03	4.5E-03	9.6E-03	8.9E-03
RMSE	0.02	0.02	0.04	0.04
R_2	0.41	0.44	0.42	0.49
R_{adj^2}	0.32	0.36	0.34	0.41

Conclusions

Two sets of methane adsorption data for shale samples were extracted from published literature, covering both original and compressively deformed conditions within a pressure range of 0–15 MPa at 30°C. The data indicate that deformation negatively affects adsorption, which may in turn influence the selectivity of the adsorption isotherms. Considering this, this study assessed theoretically CH₄ adsorption on the shale using analytical adsorption isotherm models. Based on the adoption of established adsorption isotherm models (i.e., Langmuir, Toth, and Henry), the ability of each model to predict CH₄ adsorption onto shale at subsurface conditions (15 MPa, 30°C, and moisture) was tested. The results show that the Langmuir and Tóth isotherms both provide good fits to the data, while the Henry isotherm demonstrates poor performance, as indicated by higher RMSE values and lower R² values. This reaffirms the continued relevance of classical isotherm models for describing gas–rock interactions in shale formations and highlights the importance of selecting appropriate models for reliable reserve estimation.

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